



COVER PAGE

Mini environmental station with STM32

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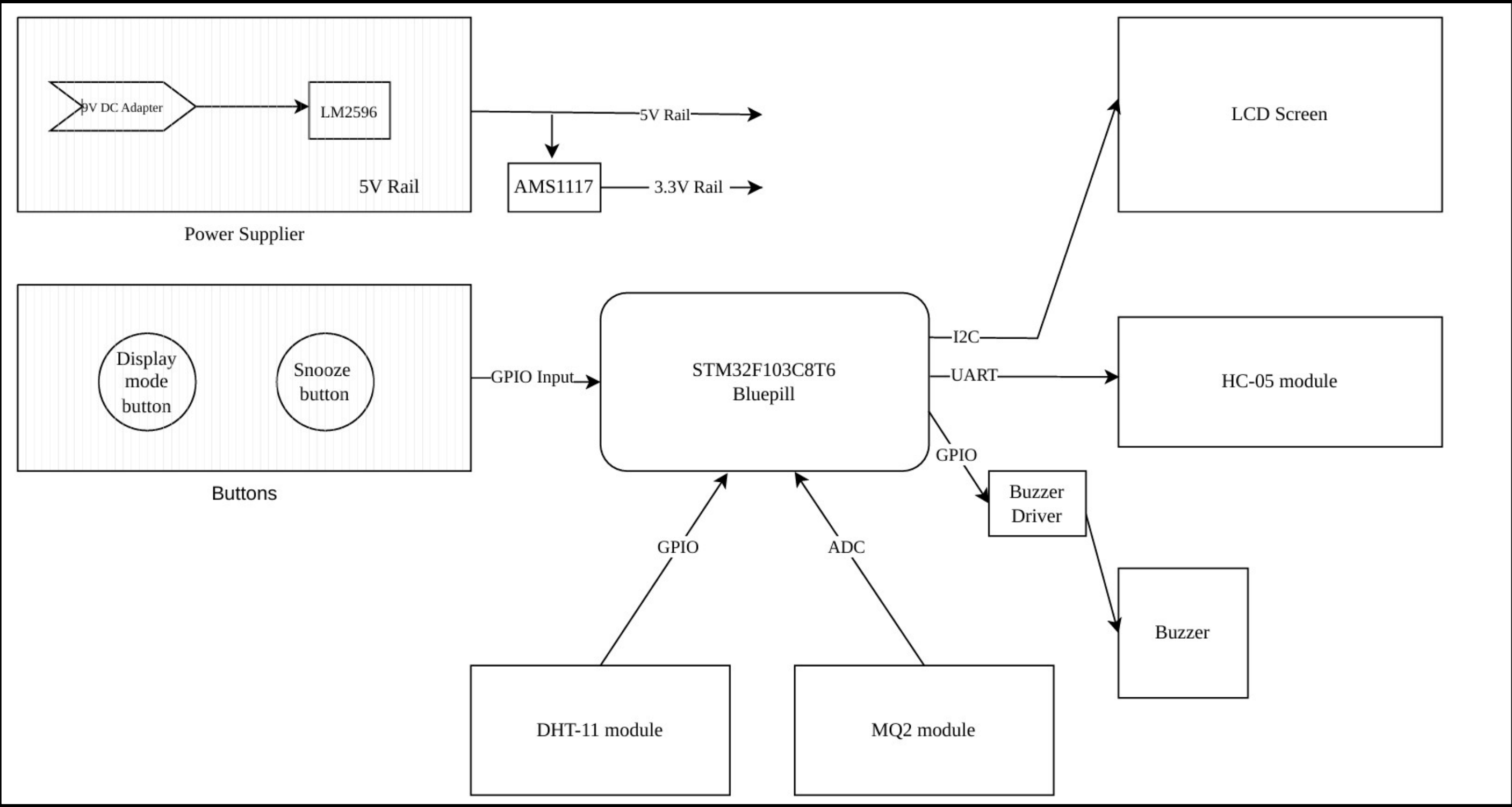
CONTEXT

CONTEXT

CONTEXT



BLOCK DIAGRAM



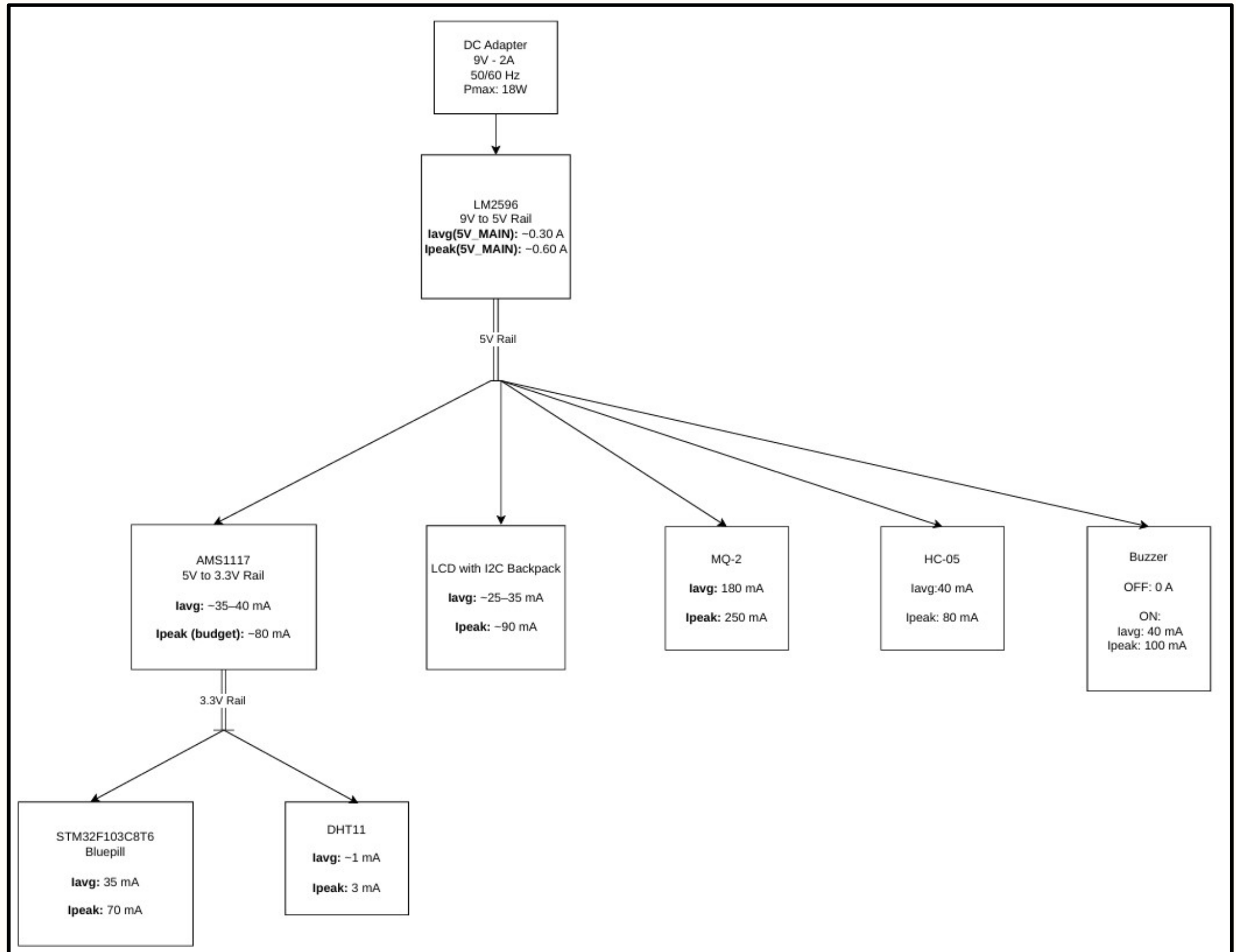
POWER TREE

LM2596 Thermal Check

Case	Iout	Pout	Ploss ($\eta=80-85\%$)	ΔT est. (θJA 40-80)	Tj @ Ta=25°C	Tj @ Ta=60°C
Average	0.30 A	1.50 W	0.26-0.38 W	11-30 °C	36-55 °C	71-90 °C
Peak	0.60 A	3.00 W	0.53-0.75 W	21-60 °C	46-85 °C	81-120 °C

AMS1117 Thermal Check

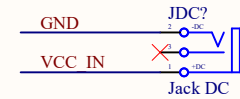
Case	Iout	Pout	Ploss	ΔT est. (θJA 50-100)	Tj @ Ta=25°C	Tj @ Ta=60°C
Nominal	80 mA	0.264 W	0.136 W	7-14 °C	32-39 °C	67-74 °C





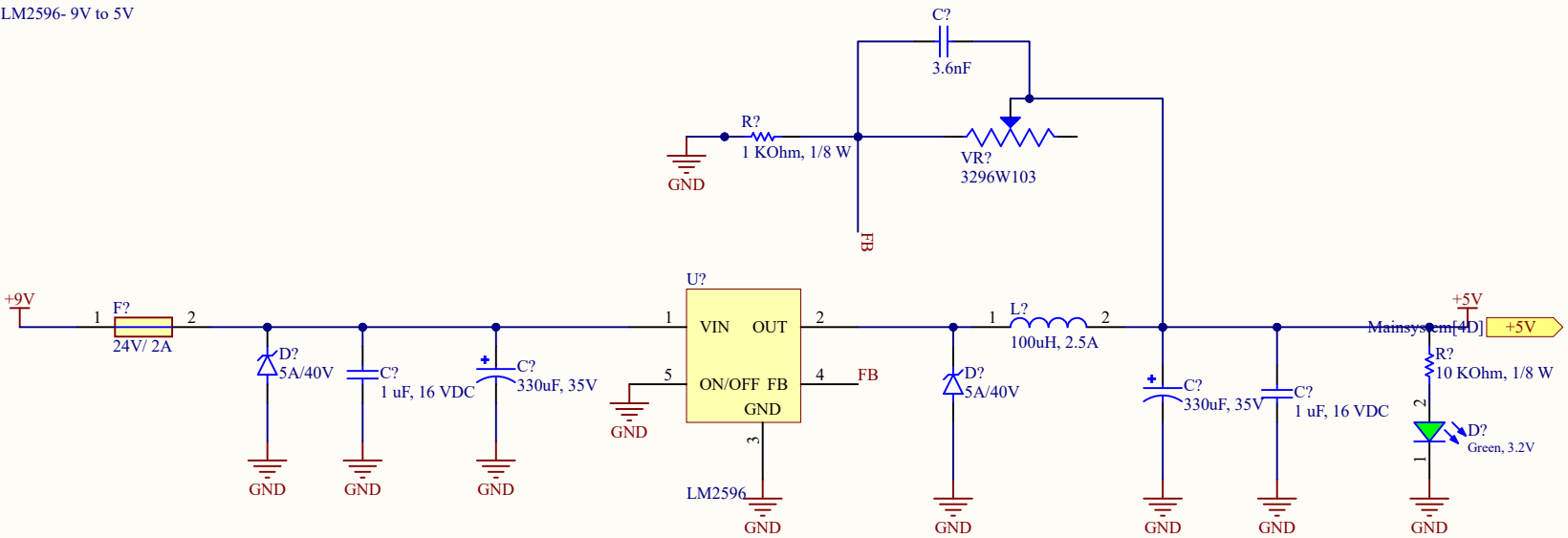
POWER SUPPLY

DC ADAPTER



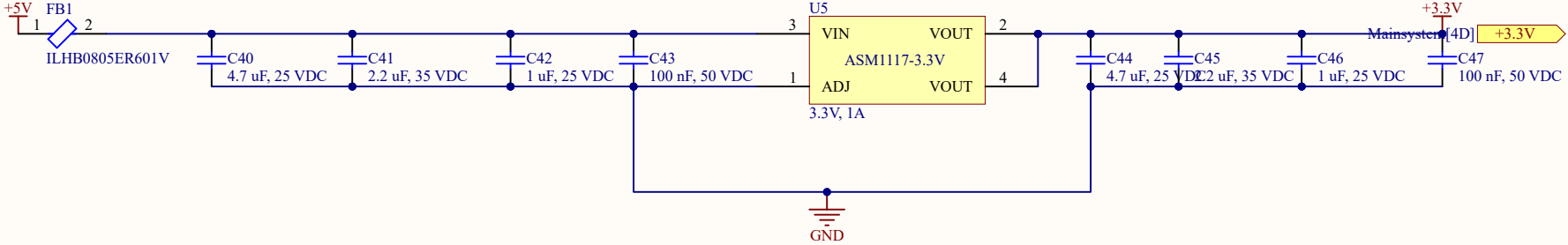
DC CONVERTER

LM2596- 9V to 5V



BUCK LDO

AMS1117- 5V to 3.3V

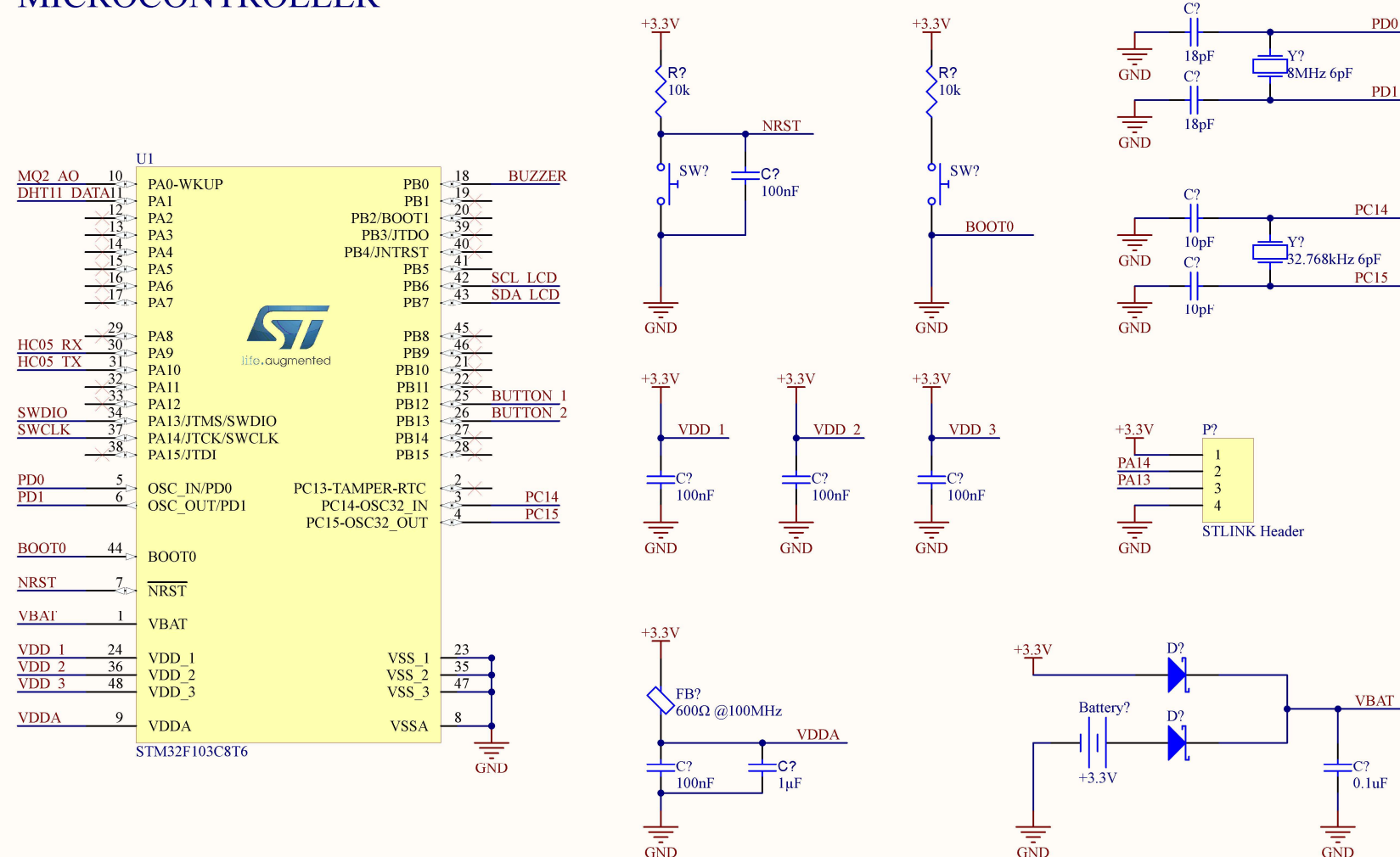


Title				Mini environmental station with STM32	
Size	Number		Revision		
A3	Embedded System Design - Group 8		MSc. Bui Quoc Bao		
Date:	3/07/2026		Sheet of	4 of 5	
File:	C:\Users\...\Powersupply.SchDoc		Drawn By:	Pham Hong An	

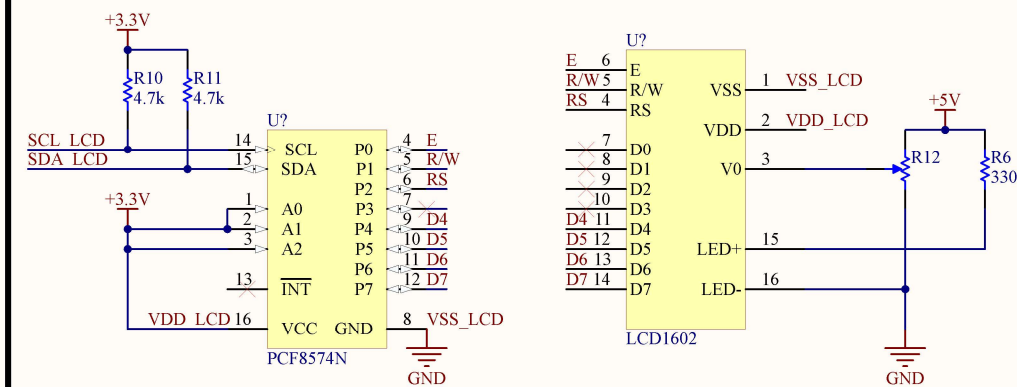


MAIN SYSTEM

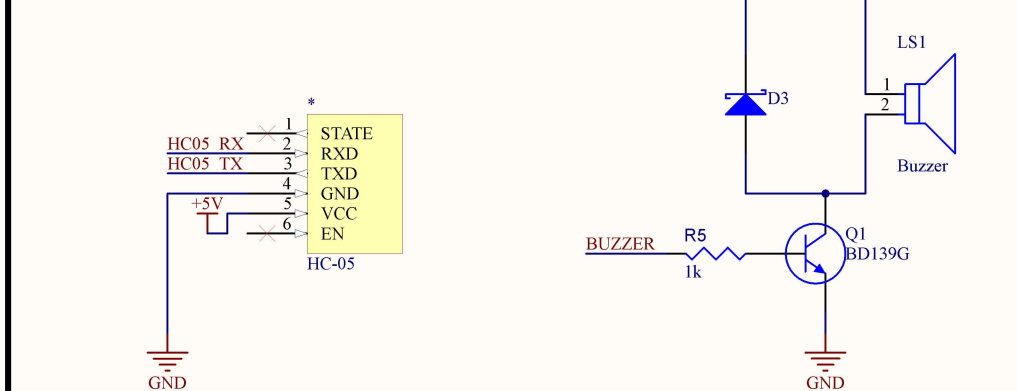
MICROCONTROLLER



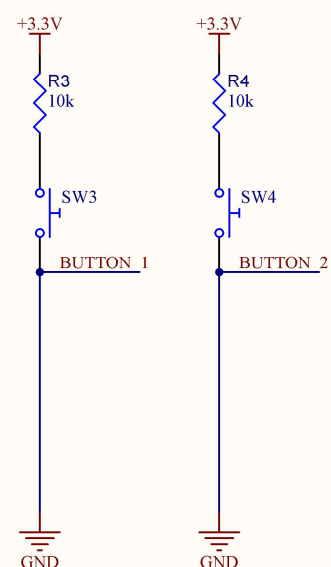
LCD DISPLAY



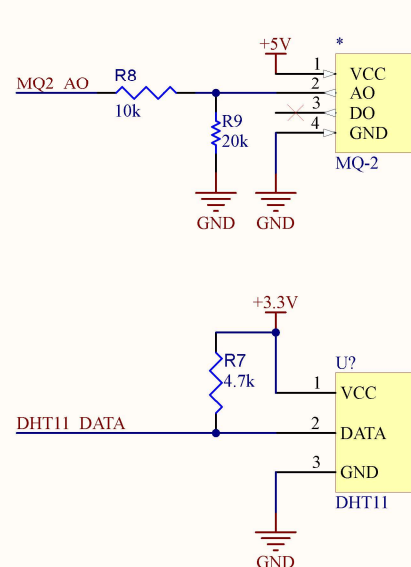
COMMUNICATION & ALERT



BUTTONS



SENSORS



POWER RAILS

